

PHYSICAL WORLD 10 PROJECT SC02 TEAM 01

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OUR MECHANISM

1. **Rubber band-powered mechanism** - Rubber bands store potential energy when twisted or stretched. Upon release, elastic potential energy is converted to kinetic energy that propels the car forward. The method is chosen due to its accessibility, cost-effectiveness, and ease of use compared to other techniques. Additionally, rubber bands give more power to car because of its elasticity that allows large deformations
2. **Larger wheels** - The car uses larger wheels to generate more torque ($\tau = F \times D$) to overcome obstacles like bumps and gaps while also increasing ground clearance and approach angles for better handling.
3. **Wheels friction** - covering wheels with rubber bands for better grip against surface
4. **Shifting of center mass** - White load to allow even weight distribution for car without weights



INITIAL CAD DESIGNS



3D-printed PLA - All car parts are fabricated using 3D-printed PLA from the Fab Lab, except for the ball bearings and rubber bands. PLA was chosen as the primary material due to its ease of fabrication, lightweight properties, and relative durability.

Design challenges :

1. **Wheels design** - The wheels were too small and did not give enough torque to turn. Moreover, the axle diameter was smaller than the axle hole on the wheels, making the axle spin without the wheels spinning. The Wheels also kept sliding over from the axle due to a lack of secure attachments between the wheels and the axle.
2. There was no **ground clearance** for the rubber band mechanism.
3. **The car body was too thick and heavy**, making the car move slower.

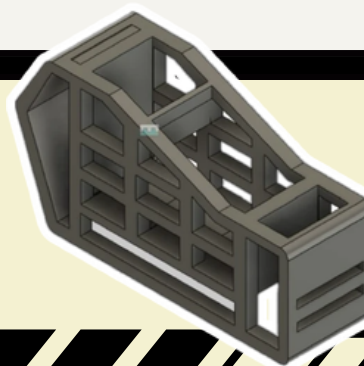
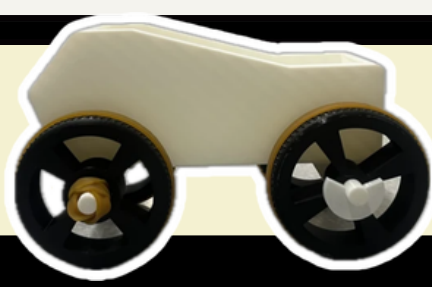
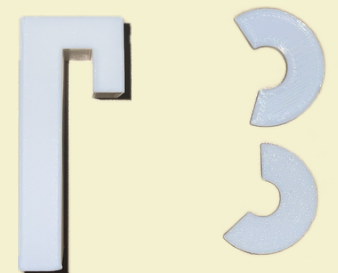
INITIAL PROTOTYPE

FINAL DESIGN

Several modifications done to solve problems:

1. **Modified wheels design** - making bigger wheels, remodeling the middle hole connected with axle is using a square shape to allow wheels more with axle, making wheel surface smoother to allow rubber band coverage.
2. **C-clips(washer)** - preventing wheels from slipping off axle
3. **Increasing the height of the triangular axle** for better ground clearance to attach rubber band mechanism
4. **Redesigning chassis** - making the chassis weigh lesser. We realized this design shifts the center of mass to the back of car because its heavier, hence we added white load to balance the center of mass

white load Washers



LEARNING POINTS

The rubber band mechanism of the toy car delivers an initial burst of energy, which presents both advantages and challenges. This energy surge is beneficial for climbing ramps and overcoming obstacles. However, it also necessitates an optimal weight for the car. If the car is too lightweight and lacks sufficient wheel grip, it may topple backward due to the sudden force. By observing and comparing with other groups, we noted that some opted for a mousetrap mechanism, which provides a more consistent and stable energy output. The energy released in their design is proportional to the extension of the spring mechanism, resulting in improved stability and control of the car.